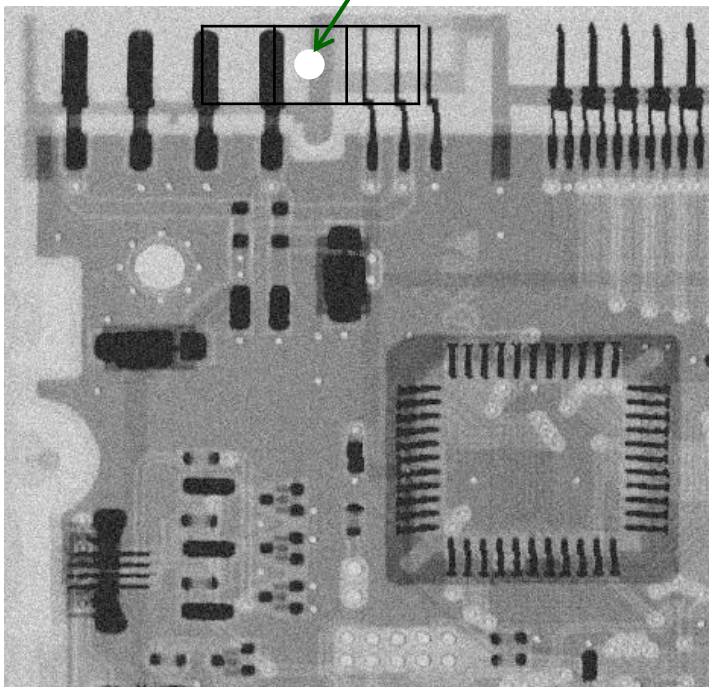
Assignment : Denoising (1/4)

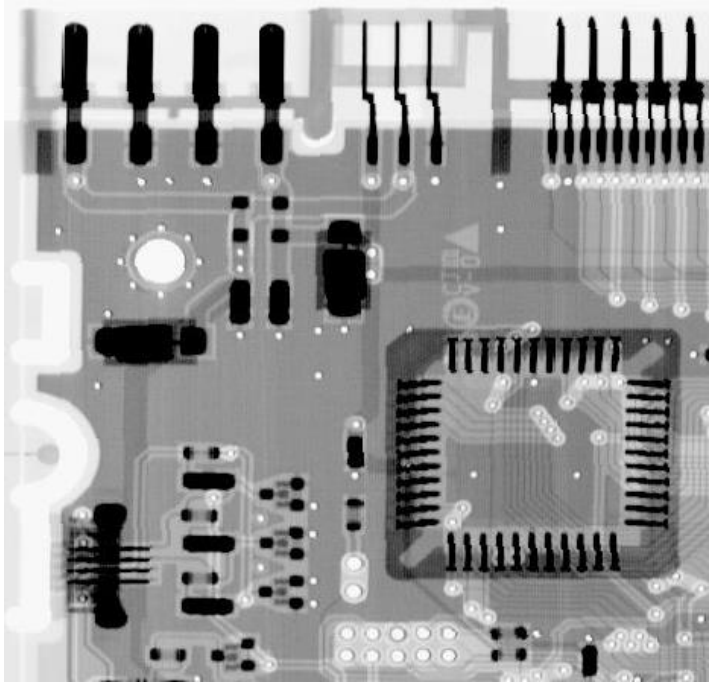
- **Why do we learn about sorting ?**
  - **Example : noise elimination in image processing**

    * Take the median value at every pixel position as follows :

Current pixel ➔ **Take the median among three values !**



Noisy image



Reconstructed image

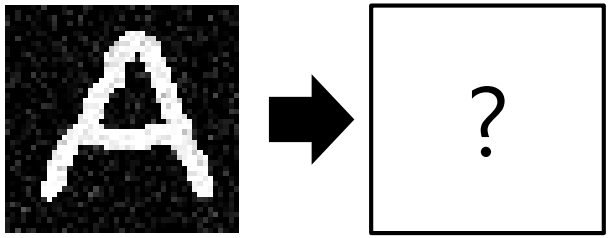
Assignment : Denoising (2/4)

- **Implementation example**
  - **Download "noisy_data.txt" from the class webpage**
  - **You can read that file using the following function**

```
void read_file(int *input, int height, int width, char filename[])
{
    int x, y;
    FILE *fp;

    fp = fopen(filename, "r");

    for(y = 0; y<height; y++){
        for(x = 0; x<width; x++){
            fscanf(fp, "%d", &input[y*width+x]);
        }
        fscanf(fp, "₩n");
    }
    fclose(fp);
}
```



Assignment : Denoising (3/4)

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▪ Implementation example – cont'd

- **Download "noisy_data.txt" from the class webpage**

```
#include <stdio.h>
#include <stdlib.h>

void read(int *, int, int, char *);
void median_filtering(int *, int *, int, int);

void main()
{
    int height = 44, width = 45; // image width and height
    ----------------------------------------------------------------
    |                    Make your code !                          |
    ----------------------------------------------------------------
}
```

➡ **Check your result using the excel program !**

Assignment : Denoising (4/4)

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▪ Implementation example – cont'd

- **구현 참고 사항**

1) 모든 위치에서 W x W 크기의 윈도우 영역을 정의하여 Sorting 수행
   → W 값은 코드에서 설정할 수 있도록 할 것
2) Median Filtering 수행 : 현재 위치의 값을 1)의 결과의 중간값으로 대체
3) W x W 크기의 윈도우 설정 시, Out of Boundary 처리 해줄 것
4) void median_filtering(int *, int *, int, int);
   → void median_filtering(입력 데이터, 출력 데이터, height, width)
5) Median Filtering을 수행한 결과는 새로운 Array에 저장할 것 (즉, 입력 데이터에 값을 덮어 쓰면서 수행하지 말 것)
6) Sorting 알고리즘은 배운 내용 중 원하는 알고리즘 사용